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Optimising crop calendars with management practices promotes climate-smart agriculture in wheat-maize rotations of the North China Plain

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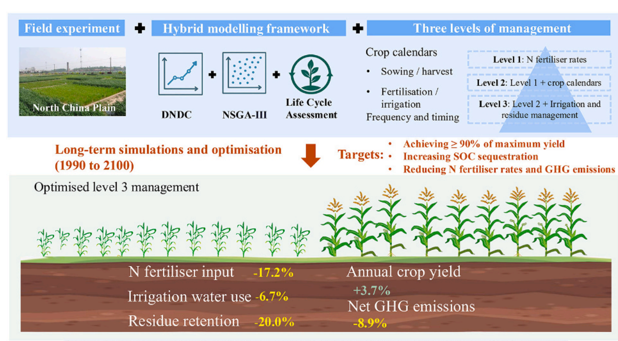
HIGHLIGHTS

- Optimised crop calendars increased yields while reducing N use and GHG emissions.
- Incorporating irrigation and residue retention into optimisation further reduced emissions.
- Climate-adaptive crop calendars alleviate maize yield losses under warming.
- Soil carbon sequestration and lower inputs drive long-term GHG mitigation potential.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Optimising crop calendars by adjusting sowing dates and the timing and frequency of key management practices can enhance crop productivity while reducing greenhouse gas (GHG) emissions. However, limited research has explored how farmers dynamically adapt crop calendars and practices in response to climate shifts to support climate-smart agriculture.
OBJECTIVE: This study developed a DNDC-based hybrid modelling framework to evaluate adaptive management strategies for supporting climate-smart agriculture under future climate scenarios.
METHODS: We assessed three management levels: fertiliser application rates (level 1); fertiliser rates combined with crop calendar adjustments, including fertiliser timing, frequency, as well as sowing and harvesting dates (level 2) and level 2 plus irrigation and residue retention (level 3). The framework was designed to optimise management under multiple objectives, including increasing crop yield, SOC sequestration, while simultaneously reducing N input and GHG emissions.
RESULTS AND CONCLUSIONS: From 1990 to 2100, the optimised crop calendars were identified: delaying wheat basal fertilisation (+5 d,) while advancing top-dressing (−5 d), postponing wheat sowing (+5 d) and advancing

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maize sowing (−9 d); advancing both fertilisation events in maize (−9 d, −3 d); aligning irrigation with fertilisation; and adding one irrigation event during the maize bell stage. Compared with historical practices, these adjustments increased annual crop yields and NUE by 4.2 % and 15.8 %, respectively, while reducing net GHG emissions and GHG intensity by 5.1 % and 8.5 %, respectively. The optimised management reduced N inputs, irrigation water and residue retention by 17.2 %, 6.7 % and 20.0 %, respectively.

SIGNIFICANCE: These findings demonstrate that adaptive crop calendars can significantly advance climate-smart agriculture and should be incorporated into climate change impact assessments.

1. Introduction

Nitrogen (N), particularly in the forms of synthetic fertilisers, is crucial for increasing food production to sustain the growing global population (Foley et al., 2011; Gerten et al., 2020). However, in some intensive agricultural regions, excessive N use has reduced N use efficiency and increased N loss to the environment, causing non-point pollution and greenhouse gas (GHG) emissions (Gu et al., 2023; Zhang et al., 2015a). These challenges hinder the realisation of climate-smart agriculture, which seeks to ensure food security by maintaining high agricultural productivity while simultaneously addressing climate change by reducing GHG emissions (FAO, 2013). Implementing effective agricultural management practices (e.g. fertilisation, irrigation, residue management) have the potential to achieve these climate-smart goals (Xia et al., 2017). However, these practices must be tailored to specific sites, according to local climate and environmental conditions. Identifying the site-specific optimal practices is critical but challenging.

Agricultural practices, interacting with local climatic and edaphic conditions, determine the yield formation and associated GHG emissions, yet generally fail to achieve a win-win outcome. For instance, high yields often depend on intensive fertiliser use, which—if mismanaged—can increase GHG emissions (Yu et al., 2019). Similar trade-offs occur in residue retention, which can enhance soil fertility and soil organic carbon (SOC) through providing C inputs (Liu et al., 2023; Wang et al., 2019) but may also alter the soil stoichiometric balance and increase nitrous oxide (N_2O) emissions (Amelung et al., 2020; Fang et al., 2018). These examples highlight the potential of optimised management practices to achieve climate-smart outcomes (Gu et al., 2023). Despite this potential, management optimisation is constrained by the lack of sufficient information on crop calendars (e.g. fertilisation and irrigation times, sowing and harvest dates). Adjusting crop calendars can align crop growth with favourable climatic conditions, enhance nutrient and water synchronisation, and reduce input losses (Bonelli et al., 2016; Hunt et al., 2019; Tsimba et al., 2013). The potential for incorporating crop calendars into management optimisation remains unclear. Besides, recent studies show that climate change has accelerated crop phenology progress, highlighting the growing importance of climatic factors in determining yield and GHG emissions (Menzel et al., 2006; Minoli et al., 2022; Yang et al., 2024). To date, few studies have assessed how farmers dynamically adapt crop calendars and management practices in response to climate shifts.

Identifying optimal solutions through field-based experiments is inherently challenging, as the full range of practice combinations or precise application levels (e.g., fertilisation and irrigation rates) cannot be comprehensively assessed due to cost constraints, practical limitations, and the limited duration of studies. Process-based models have been widely used to simulate agricultural biogeochemical processes under diverse management scenarios (Jiang et al., 2025; Zhao et al., 2023). While these models are capable of simulating a large number of scenarios following careful calibration, evaluating trade-offs among output variables (particularly when more than three competing objectives) remains a significant challenge. Owing to its capacity to handle high-dimensional data, machine learning facilitates the exploration of trade-offs and the identification of optimal solutions that are often difficult to achieve using traditional modelling approach. Recently, the integration of machine learning approach with process-based models has recently made identification of optimal agricultural practices possible (Lam et al., 2024; Yan et al., 2024). This integration enables comprehensive assessment of various combinations of management practices, which is crucial for developing practical, scalable solutions tailored for local implementation.

The North China Plain (NCP) is a critical region for the world's total wheat and maize production. However, unsustainable management practices has made this area one of the global hotspots for GHG emissions (Wang et al., 2024a). Although numerous field- and model-based have explored management optimisation in this region, they have overlooked the potential of crop calendar scenarios or the diversity of management practices or the integration of future climate-adaptive strategies (Li et al., 2016; Wang et al., 2018; Wang et al., 2012; Zhao et al., 2015). Therefore, a systematic understanding of the potential for optimising crop calendars and management practices to achieve climate-smart agriculture remains largely lacking. In this study, we used the process-based DNDC model to simulate seasonal crop yields, N_2O emission, and annual SOC stocks from 1990 to 2099. We then applied multi-objective optimisation using the NSGA-III algorithm to identify the optimal practices for achieving climate-smart goals. A reference scenario representing historical practices, assuming no action would be taken in response to climate change, was compared with fine-tuned scenarios incorporating three levels of adjustment in crop calendars, fertilisation, irrigation, and residue retention. The aims of the study are to: (1) identify the optimal crop calendars and management practices to achieve climate-smart agriculture; (2) quantify the potential improvement in crop yields and the reduction in GHG emissions and N inputs if farmers adapt the optimised practices in response to climate change.

2. Methods

2.1. The study site and experimental design

A long-term field experiment with winter wheat *(Triticum aestivum* L.*)* and summer maize *(Zea mays* L.*)* double-cropping system was conducted from June 1990 to June 2015 at the Yucheng Comprehensive Experiment Station (36°51′N, 116°34′E) in the NCP. The experimental site is typical of the NCP, characterised by long-term intensive agriculture (Meng et al., 2016), with a high annual N input of approximately 480 kg N ha^{-1} per year to maintain high wheat and maize yield (5.98 and 8.07 Mg ha^{-1} per year, respectively) from 1990 to 2015. During the study period, the annual mean temperature was 13.7 °C, and the annual mean precipitation was 600 mm. The soil was classified as Calcaric Fluvisol according to the FAO-UNESCO system. The initial topsoil (0-20 cm) is silty clay loam, contained 23 % clay, with a pH of 7.1, a bulk density of 1.35 g cm^{-3} and a soil organic carbon (SOC) content of 5.39 g kg^{-1}. The details of local agricultural management (historical practices) are shown in Table S1.

Grain yields (kg ha^{-1}) for wheat and maize were recorded after air drying to 13 % moisture. Soil samples (0–20 cm) were collected annually at the harvest of winter wheat, with three subsamples taken at each event and mixed thoroughly to form a composite sample. After removing roots and stones, the samples were air-dried and passed through a 0.15 mm sieve for SOC determination. The Walkley–Black Chromic Acid Wet Oxidation Method was then used to determine SOC (g kg^{-1}) (Li et al., 2023; Walkey, 1934). As wheat and maize were sown in aerobic dryland soils in upland crop fields, the contribution of CH_4 to GHG emissions is minimal (in some cases even a sink) (Linquist et al., 2012). Therefore, we considered nitrous oxide (N_2O) as the primary source of soil GHG emissions in this study. N_2O fluxes (μg N ha^{-1} per hour) was measured monthly using the active-passive closed chamber system (Tu and Li, 2017) in 2003, 2016 and 2017. N use efficiency (NUE) (kg grain kg^{-1} N^{-1}) is calculated as crop yield per unit of N fertiliser applied, while greenhouse gas (GHG) intensity (kg CO_2-eq kg^{-1} N^{-1}) as GHG emissions per unit of N fertiliser applied.

2.2. Climate data

The historical climate data, including maximum, minimum and average temperature, as well as precipitation, were observed and recorded daily since 1990 at the meteorological station of the Yucheng Comprehensive Experiment Station. Future daily climate from present to 2100 was projected by six Global Climate Models (GCMs) (detailed descriptions in Table S2). These GCMs were sourced from the Coupled Model Inter-comparison Project phase 6 (CMIP6) under two Shared Socioeconomic Pathways (SSPs) (Arias et al., 2021). The two SSPs used in this study were SSP2–4.5 (SSP245) and SSP5–8.5 (SSP585), representing medium and high-end radiative forcing scenarios, respectively (O'Neill et al., 2016). We first extracted the climate outputs for the study site from the GCMs, and then used linear scaling method to bias-correct the daily maximum and minimum temperatures and the quantile mapping method for precipitation (Gudmundsson et al., 2012; Wood et al., 2004). To enhance reliability and reduce uncertainties in the multi-model projections, the multi-model ensemble approach was employed, taking an unweighted average of outputs from six GCMs (Feng et al., 2019; Wang et al., 2020). Corrected projections (Fig. S1) for the period of 2020 to 2100 indicate that the average temperature will increase by 31.8 % and 45.2 % in the maize growing season and by 10.2 % and 32.2 % in the wheat growing season, with precipitation in the maize season increasing by 1.8 and 1.9 times, but decreasing in the wheat season by 43.6 % and 7.9 %, under SSP245 and SSP585, respectively.

2.3. The DNDC model

The DNDC model is a process-based biogeochemical model developed for the simulation of the biological processes in agroecosystems (Li et al., 1992). It simulates crop growth and yield on a daily time step, driven by potential water and N demands, which are determined by crop physiological parameters (e.g. maximum biomass, shoot/root partitioning, C/N ratio, thermal degree days) and environmental conditions such as radiation, temperature, precipitation. The resulting biomass subsequently influences soil water content, SOC and N pools, thereby influencing all biogeochemical processes. SOC decomposition is represented using first-order kinetics across multiple SOC pools, with rates regulated by soil moisture, temperature and aeration status. N transformations, including urea hydrolysis, nitrification, denitrification, volatilisation, plant uptake, and immobilisation, are simulated as functions of environmental and substrate conditions (Giltrap et al., 2010; Li, 2000). DNDC has been verified and widely used in numerous studies on the responses of wheat–maize cropping systems to environmental, climatic and management variations in the NCP (Zhang et al., 2017, 2024; Zhang et al., 2015b).

2.3.1. Calibration and validation

We used the historical observations from 1990 to 1994 to calibrate the crop parameters of wheat and maize in DNDC (version 9.5). The calibrated parameters included maximum biomass; the allocation fraction of grain, leaf and stem; C:N ratios of grain, leaf, stem and root; as well as crop-specific thermal degree requirements and water demand. Then we run the calibrated DNDC model for the period of 1994–2014, and compared the DNDC-simulated yields, SOC and N_2O emissions with observed values for model validation. The normalized root mean square error (NRMSE), Nash-Sutcliffe efficiency (NSE) and index of agreement (d) were used to quantify and assess the model performance (Eqs. (1–3)) (Jiang et al., 2023).

$$NRMSE = \frac{\sqrt{\sum_{i=1}^{n}(S_i - O_i)^2}}{\overline{O}} \times 100 \tag{1}$$

$$NSE = 1 - \frac{\sum_{i=1}^{n}(S_i - O_i)^2}{\sum_{i=1}^{n}(O_i - \overline{O})^2} \tag{2}$$

$$d = 1 - \frac{\sum_{i=1}^{n}(S_i - O_i)^2}{\sum_{i=1}^{n}(|S_i - \overline{O}| + |O_i - \overline{O}|)^2} \tag{3}$$

where S_i, O_i are the simulated and observed values, respectively; $\overline{O}$ is the average observed values; and n is the number of observations. The details of performance categories are shown in Table S3.

2.4. Estimating net GHG emissions

A life cycle assessment (LCA)-based approach (Liu et al., 2021; Xiao et al., 2024) was used to estimate the net GHG emissions, consisting with three main components: (1) SOC sequestration (ΔSOC), (2) GHG emissions from production, including fertiliser production, transportation, irrigation and combustion of fossil fuels, and (3) direct N_2O emissions (emissions from the soil) and indirect N_2O emission caused by fertilisation (NH_3 volatilisation and NO_3^-leaching). This comprehensive assessment allowed us to evaluate the contributions of specific practices to promoting climate-smart agriculture. The Eqs. (4–7) for estimating net GHG emissions (kg CO_2-eq ha^{-1} per year) are as follows:

$$Net\ GHG_{emission} = Input_{emission} + N_2O_{emission} + C_{emission} \tag{4}$$

where

$$Input_{emission} = N_{input} \times EF_{Ninput} + Irr \times 9.2 \times EF_{Irr} + SR \times (EF_{Chopping} + EF_{Incoporation}) \tag{5}$$

$$N_2O_{emission} = \frac{44}{28} \times GWP \times \left(N_2O_{DNDC} + N_{input} \times 0.1 \times EF_{Nvol\_N_2O} + N_{input} \times 0.3 \times EF_{Nleach\_N_2O}\right) \tag{6}$$

$$C_{emission} = -\Delta SOC \times \frac{44}{12} \tag{7}$$

where $Input_{emission}$ (kg CO_2-eq ha^{-1} per year) are the GHG emission from agricultural inputs, including N fertiliser production (cradle to gate), power consumption for irrigation, straw chopping and incorporation. N_{input} (kg N ha^{-1}), Irr (mm), SR (%) are the N input rates from N fertilisers, irrigation amount and the fraction of straw retained. 9.2 (kWh^{-1} mm^{-1}) is the electricity consumption by irrigation (Xiao et al., 2024). EF_{Ninput}, EF_{Irr}, $EF_{Chopping}$ and $EF_{Incoporation}$ are the GHG emission factors of N fertiliser production, electricity consumption for irrigation, straw chopping for maize and straw incorporation at a depth of 20 cm, respectively. In this study, these factors are 8.3, 1.23, 16.58 and 22.29, respectively, as sourced from ref. (Li et al., 2016; Li et al., 2019; Liu et al., 2021; Xin and Tao, 2021). $N_2O_{emission}$ (kg CO_2-eq ha^{-1} per year) is the total N_2O emission, including both direct and indirect N_2O emissions; 44/28 is the conversion rate of N to N_2O; GWP is the global warming potential of N_2O over 100 years, which is 273 according to the Sixth Assessment Report (AR6) of the Intergovernmental Panel on Climate Change (IPCC) (Nabuurs et al., 2022). N_2O_{DNDC} (kg N ha^{-1}) is the annual cumulative DNDC-simulated N_2O emissions. $EF_{Nvol\_N_2O}$ and $EF_{Nleach\_N_2O}$ are the N_2O emission factors of NH_3 volatilisation and NO_3^- leaching, respectively. In this study, these factors are 0.01 and 0.0075, as sourced from ref. (Intergovernmental Panel on Climate Change, 2006). 0.1 and 0.3 are the conversion rate of total N input to volatilized and leached N (Intergovernmental Panel on Climate Change, 2006). $C_{emission}$ (kg CO_2-eq ha^{-1} per year) are the carbon emissions. 44/12 is the conversion rate of C to CO_2.

2.5. DNDC–NSGA–III hybrid model

2.5.1. Management practices at three levels

Based on technical thresholds and local farmer acceptance, management practices and crop calendars were grouped into three levels (Gu et al., 2023): adjusting (1) N fertiliser application rates; (2) N fertiliser application rates and crop calendars, including the application dates, sowing and harvest dates; and (3) N fertiliser application rates, crop calendars and irrigation and residue management. Then, we generated scenarios of fine-tuned combinations for each level. For level 1, we established gradient N application scenarios for both wheat and maize with N application rates ranging from 0 to 400 kg N ha^{-1} per year, in intervals of 25 kg N ha^{-1}. For levels 2 and 3, we randomly generated fine-tuned 10,000 combinations of the practices within each level according to the constraint conditions in the Supplementary Methods. The DNDC was then run for the period of 1990 to 2100 to simulate wheat yield, maize yield, N_2O emission and ΔSOC for these scenarios.

2.5.2. Multi-objective optimisation

In this study, we developed a multi-objective optimisation framework integrating the DNDC model with the Non-dominated Sorting Genetic Algorithm III (NSGA–III) (Yan et al., 2024) The NSGA–III, grounded in the Edgeworth-Pareto principle, employs a genetic algorithm incorporating non-dominated sorting and reference point mechanisms to simultaneously optimise multiple objectives (typically >3) (Deb and Jain, 2014). This algorithm identifies a set of optimal solutions, known as the Pareto front, which represents the best achievable trade-offs among the specified objectives (Zitzler and Thiele, 1999).

The DNDC-generated scenario outputs for each decade until 2100 served as the parental population, which were evaluated based on the non-dominated sorting rankings and subsequently used to compare potential solutions. The NSGA–III was run a total of 36 times, encompassing two-level practices, two climate change scenarios, and nine time periods (2 × 2 × 9). Pareto fronts for each simulation were generated through 10 iterations per run, considering four optimisation objectives: maximum wheat yields, maximum maize yields, maximum SOC and minimum N_2O emission (Fig. S3). The optimal practices were then selected from the Pareto fronts based on the targets of climate-smart agriculture, specifically aiming to achieve at least 90 % of the maximum theoretical yields with reduced N inputs and net GHG emissions. Finally, the impact of optimal practices on yields and net GHG emissions was evaluated in comparison to scenarios where all practices are assumed to remain unchanged in the future.

3. Results

3.1. DNDC model performance for long-term simulation

The verified DNDC model demonstrated robust performance in simulating crop yields, SOC and N_2O emissions over a 25-year period (Fig. 1, Fig. S2). It achieved excellent performance in simulating wheat yield, SOC, and N_2O emissions, with d-values of 0.95, 0.96, and 0.99, respectively, and showed good performance for maize yield, with a d-value of 0.82.

3.2. Responses of yields, N_2O and SOC to nitrogen rates

We optimised N rates (level 1 practice) and found that both maize and wheat yields increase with increasing annual N inputs until they reach 90 % of their theoretical maximum yield (5632.4 kg ha^{-1} for wheat at 200 kg N ha^{-1} per year and 8620.4 kg ha^{-1} for maize at 325 kg N ha^{-1} per year) (Fig. 2). NUE initially increases with increasing N inputs, reaching maximum values of 38.3 kg grain per kg N for maize at 75 kg ha^{-1} applied per year and 25.6 kg grain per kg N for maize at 175 kg ha^{-1} applied per year, before declining with further N application. SOC increases with increasing N inputs with a maximum value of 6.3 g kg^{-1} at 225 kg N ha^{-1} per year per growing season (i.e., 450 kg N ha^{-1} per year). Beyond this level, further nitrogen inputs do not lead to additional

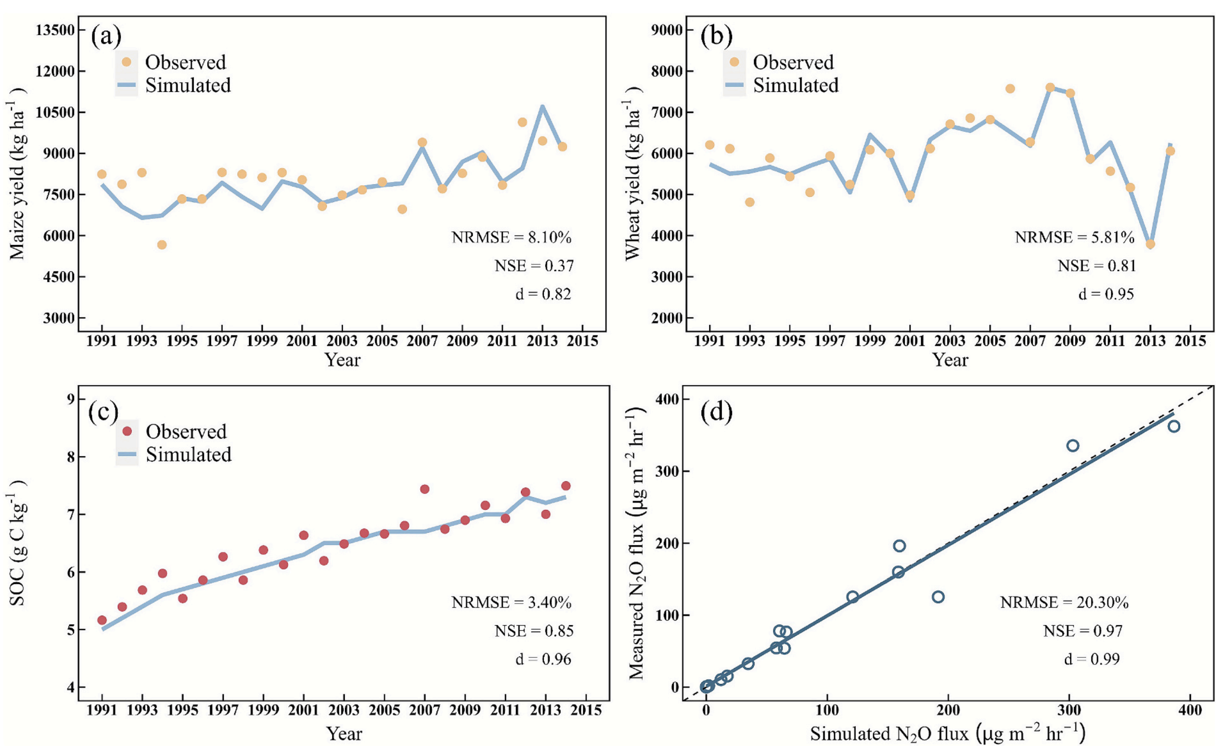

Fig. 1. Performance of DNDC model in long-term simulations of (a) maize yield, (b) wheat yield, (c) soil organic carbon (SOC) and (d) N_2O emissions.

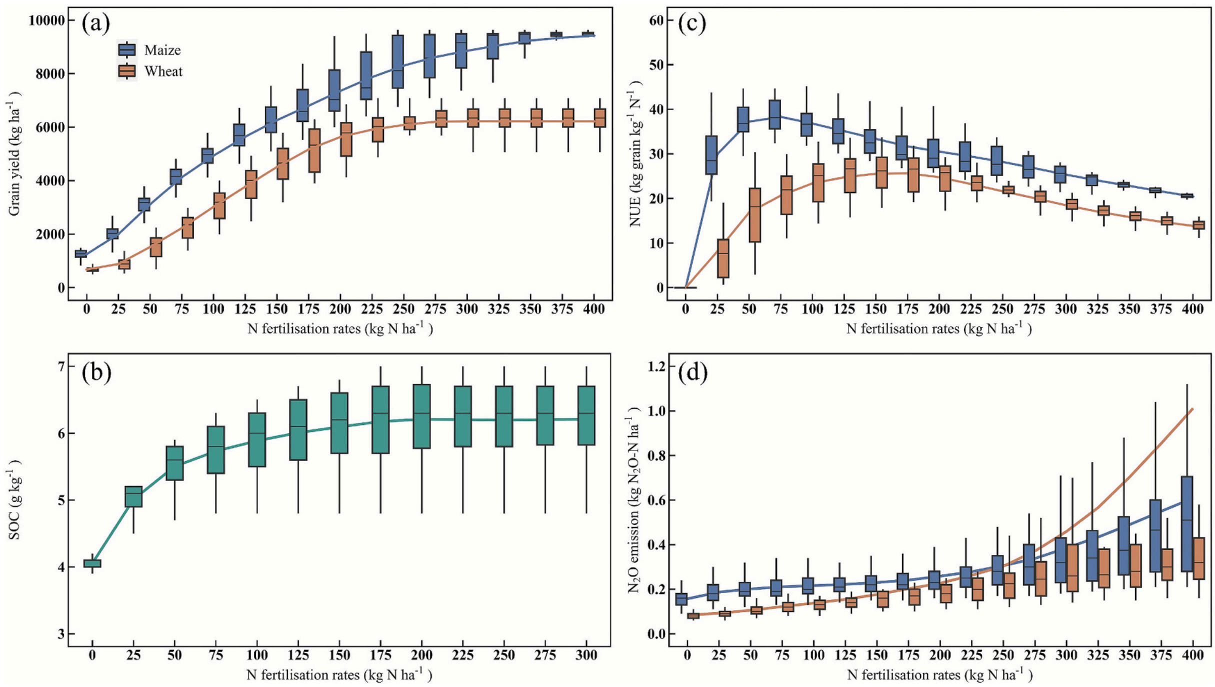

Fig. 2. Response curves for (a) crop yields, (b) soil organic carbon (SOC), (c) nitrogen use efficiency (NUE), and (d) N_2O emissions to gradient N fertilisation rates over a 30-year period. The lines in each panel are smoothed representations of the baselines, with averaged fitted values. In each box plot, the black line, two box boundaries and whiskers below and above represents median, 25th and 75th percentiles, and 10th and 90th, respectively.

SOC increases. N_2O emissions increase with increasing N during both wheat and maize seasons, showing a stronger response at higher application rates.

3.3. Optimal nitrogen management and the climate change adaptation

Optimal fertilisation practices involve reducing annual application rates, with lower and delayed basal fertilisation and earlier top-dressing in the wheat season, as well as earlier timing of both fertilisation events in the maize season. Specifically, under SSP245, total annual N inputs were reduced to 433.7 kg N ha^{-1} in level 2 practices, and further decreased to 397.4 kg N ha^{-1} when optimised through a combination of with irrigation and residue management (level 3 practices), whereas higher N (411.1 kg N ha^{-1}) were required for level 3 practices under SSP585 (Fig. 3a). Compared to conventional equal-split fertilisation between two growing seasons and between two fertilisation events per season, we found that N input should be 33.5 kg N ha^{-1} greater for maize than for wheat, and that top-dressing during the wheat season should exceed basal application by 51.0 kg N ha^{-1} (Fig. 3a).

The optimum top-dressing dates for wheat and maize should be advanced by approximately three and five days, respectively, over a 110-year period (Fig. 3d). Additionally, the optimum sowing dates for wheat should be delayed by five days, while those for maize be advanced by 10 days during the same period (Fig. 3b and c). Optimised irrigation practices indicated that annual water use could be reduced to 30.8 cm, even with an additional irrigation event at the maize bell stage, and water use during the wheat season should be 4.3 cm higher than that of the maize season (Fig. 3e). The optimum residue retention for wheat and maize were 0.71 and 0.68, respectively over the 110-year period (Fig. 3f).

3.4. Optimised yields and net GHG emissions

Under level 2 and 3 optimised practices, the annual crop yields and NUE under SSP245 are projected to increase by 5.6 % (777.3 kg ha^{-1}) and 17.9 % (5.2 kg grain ha^{-1} per kg N) compared to historical practices (Fig. 4a and b). These increases are 1.7 % (equivalent to 234.2 kg ha^{-1}) and 13.3 % (equivalent to 5.1 kg grain ha^{-1} per kg N) higher than those under SSP585. Net GHG emissions are projected to decrease by 9.4 % (767.6 kg CO_2-eq ha^{-1} per year) and by 8.4 % (678.5 kg CO_2-eq ha^{-1} per year) under SSP245 and SSP585, respectively (Fig. 4c). Additionally, GHG intensity is projected to reduce by 14.2 % (0.08 kg CO_2-eq ha^{-1} per unit yield) under SSP245, which is 9.8 % (0.06 kg CO_2-eq ha^{-1} per unit yield) higher than projected under SSP585 (Fig. 4d). Under both SSPs scenarios, compared to historical practices, level 2 practices result in increased annual yield and NUE by 576.1 kg ha^{-1} and 4.6 kg grain ha^{-1} per kg N, respectively, while reducing net GHG emissions and GHG intensity by 419.6 kg CO_2-eq ha^{-1} per year and 0.05 kg CO_2-eq ha^{-1} per unit yield, respectively. Under both SSPs scenarios, compared to level 2 practices, level 3 practices result in increased annual yield and NUE by 106.7 kg ha^{-1} and 1.1 kg grain ha^{-1} per kg N, respectively, while reducing net GHG emissions and GHG intensity by 627.6 kg CO_2-eq ha^{-1} per year and 0.046 kg CO_2-eq ha^{-1} per unit yield, respectively.

Annual crop yields under optimal practices are projected to increase from 14,380.6 kg ha^{-1} in 1990 to 14,894.3 kg ha^{-1} under SSP245 and 14,800.7 kg ha^{-1} under SSP585 by the 2070s. However, by the 2090s, yields are projected to decline to 14,550.5 kg ha^{-1} under SSP245 and a more substantial decrease to 13,265.8 kg ha^{-1} under SSP585. Net GHG emissions and GHG intensity are projected to increase from 1990 to the 2090s, with an increase rate of 129.4 kg CO_2-eq ha^{-1} per decade and 0.008 kg CO_2-eq ha^{-1} per unit yield per decade under SSP245, and 137.4 kg CO_2-eq ha^{-1} per decade and 0.016 under SSP585.

4. Discussion

4.1. Responses of yield and net GHG emissions under climate change

Current management practices are not optimal for achieving the potential maximum crop yields, NUE or minimum N_2O emissions (Fig. 2). The result indicates that increasing annual N inputs to 525 kg N ha^{-1} can achieve 90 % of theoretical maximum yields for both wheat and maize, but N_2O emissions increase exponentially. This indicates that

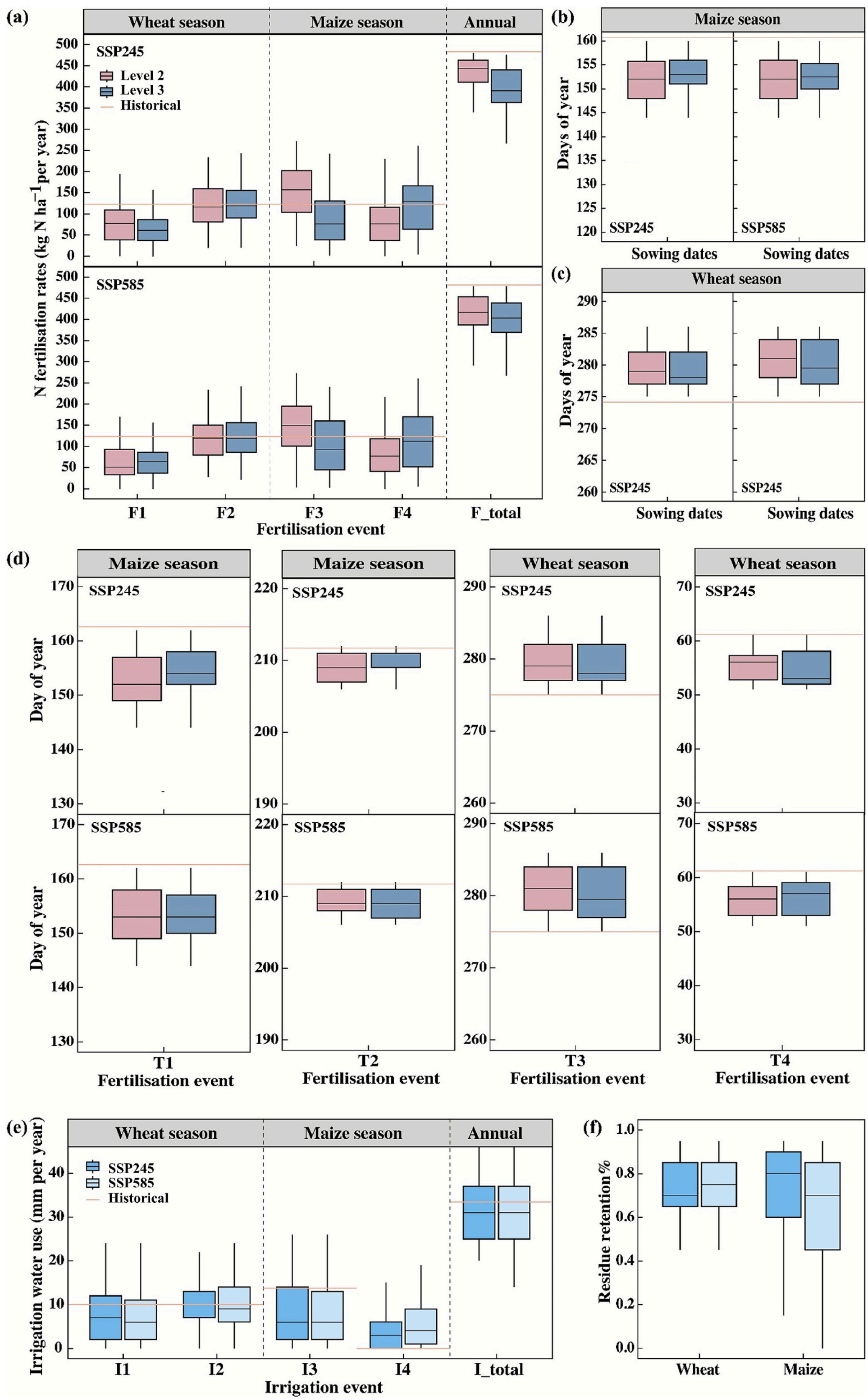

Fig. 3. Distributions of optimal N practices under the optimisation target of achieving at least 90 % maximum yields, reduced N input and net GHG emissions. a-f, (a) fertilisation rates, (b) sowing dates of maize, (c) sowing dates of wheat, (d) fertilisation application dates, (e) irrigation water use, (f) residue retention. F1 and F3 represent basal fertilisation, while F2 and F4 represent top dressing fertilisation. T1 to T4 are the dates of the four annual fertilisation events (F1-F4). The lines in each panel are values reflecting the current practice. In each box plot, the black line, two box boundaries and whiskers below and above represents median, 25th and 75th percentiles, and 10th and 90th, respectively.

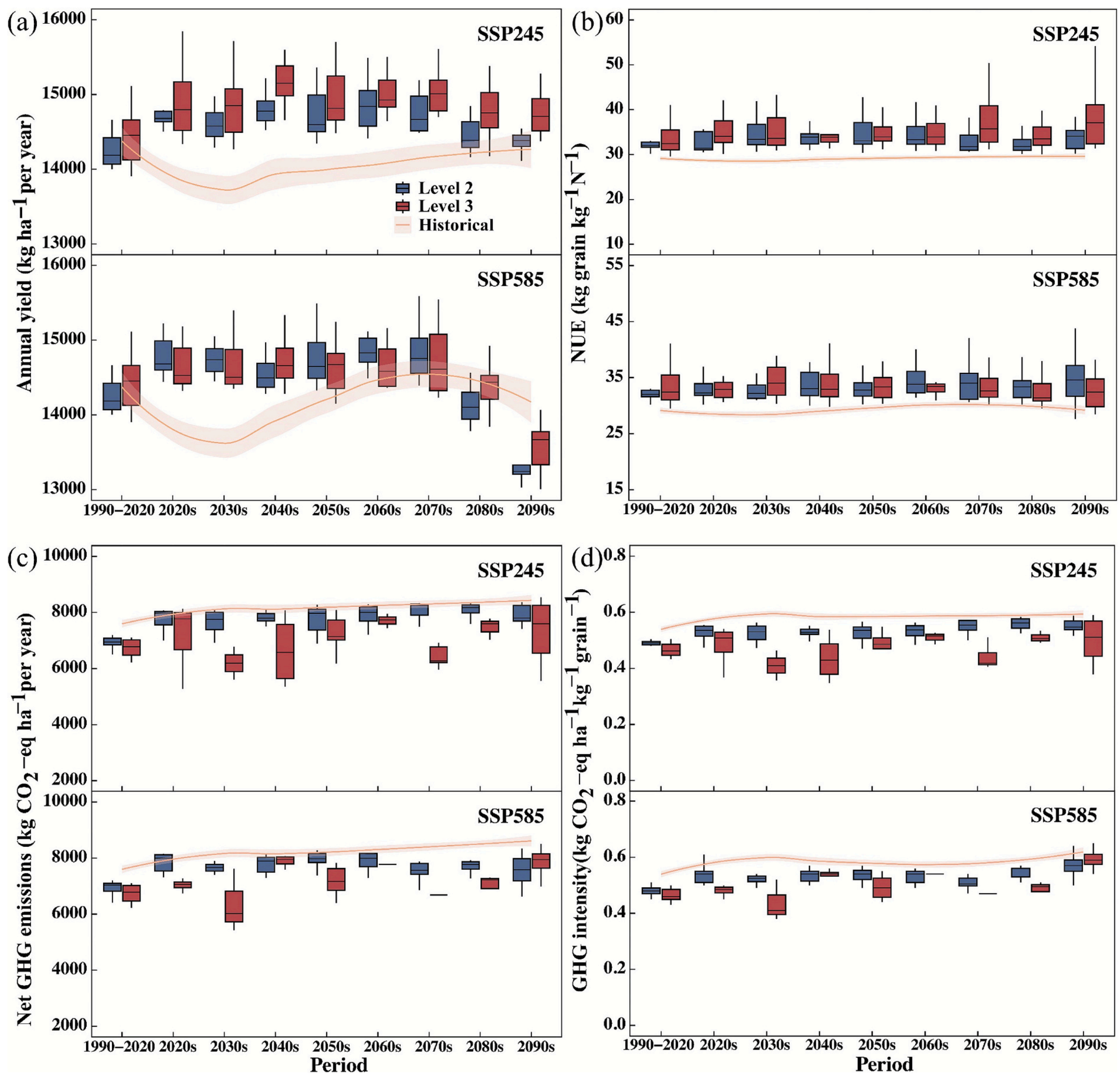

Fig. 4. Distributions of simulated (a) annual yields, (b) NUE, (c) net GHG emissions and (d) GHG intensity under current and optimised level 2 and level 3 practices from 1990 to 2100. The lines with shading (95 % confidence intervals) in each panel are smoothed representations of current practices remain unchanged throughout the period. The black line, two box boundaries and whiskers below and above represents median, 25th and 75th percentiles, and 10th and 90th, respectively.

simply adjusting N inputs (level 1 practice) is insufficient to meet these competing goals, highlighting the need for additional practices involved in the optimisation process. Compared to current practices, implementation of optimised crop calendars increases crop yields and NUE, as well as reduces net GHG emissions and GHG intensity (Fig. 4).

Over the period from 1990 to 2100, level 3 practices (adjusting fertiliser rates, irrigation, residue and crop calendars) led to only a modest 0.7 % increase in yield compared to level 2 (adjusting fertiliser rates and crop calendars), but achieved a significant 8.7 % reduction in net GHG emissions. This is likely because optimising sowing dates and key management timings in level 2 already brings yields close to their theoretical maximum, leaving limited potential for further gains through irrigation and residue management (van Ittersum et al., 2013). The significant decrease of net GHG emissions from level 2 to level 3 practices is primarily due to lower annual N inputs (Fig. 3a), as N fertiliser is the primary source of these emissions (Song et al., 2018). Additionally, in level 3 practices, optimised irrigation may help regulate soil moisture, which in turn inhibits certain soil biochemical processes associated with N_2O emissions, such as denitrification, thereby contributing to lower net GHG emissions (Pan et al., 2022). Residue retention provides additional C inputs and increases SOC, further offsetting GHG emissions (Huang et al., 2021). This suggests that even without implementing effective irrigation and residue management, improvements in yield and reductions in net GHG emissions can still be achieved as shown by level 2 practices. However, if the goal is to further reduce N inputs and GHG emissions beyond level 2, level 3 practices should be considered. Given that level 3 involves two additional interventions, careful consideration of the trade-offs between environmental benefits and the associated implementation costs is necessary.

Compared to the SSP585 (intensified fossil-fuel development), optimal management under SSP245 (sustainable and green development) is more effectively at increasing yield and NUE, while simultaneously achieving greater reductions in net GHG emissions and GHG intensity. A decline is observed in maize yields between the 2060s to 2100, whereas wheat yields remain relatively stable over the same period (Fig. S4). Similar patterns have been found in the NCP and other regions in the world (Jägermeyr et al., 2021; Xiao et al., 2024). This suggests that maize, compared to wheat in a double cropping system, is more vulnerable to future climate change. This is mainly because warmer temperatures, particularly during the summer, tend to shorten the crop's growth cycle and exacerbate water stress, which adversely impacts maize yields (Lobell et al., 2020; Zhu et al., 2018), especially when no cultivar changes are considered. Furthermore, fluctuations in precipitation contributing to yield anomalies, especially during summers marked by rainfall shortages and droughts, leading to further declines in maize productivity (Feng et al., 2022). This impact is exacerbated by the fact that farmers in this region typically either opt not to irrigate or have limited access to water resources during the maize seasons. Wheat, as a C_3 crop, generally benefited more from the fertilisation effect of rising CO_2 levels compared to maize, a C_4 crop (Ainsworth and Long, 2021). However, our results indicate that the wheat yields are likely to stabilize around the 2070s, rather than continue increasing. This suggests that the responses of yields to climate change is more crop type-dependent in the double cropping system, reflecting distinct physiological and phenological sensitivities between wheat and maize. The ambitious climate-smart goals mentioned in this study may not be sustainable in the long-term without compromising maize yields. This highlights the need for future efforts to focus on optimising maize genotypes and breeding, or carefully selecting crop types and cropping system.

The reduction in net GHG emissions achieved through the implementation of optimised practices is predominantly driven by soil C sequestration, rather than by mitigating direct soil N_2O emissions (Fig. S4). Additionally, the optimised practices reduced annual water use and N inputs (Fig. 3), thereby lowering net GHG emissions by decreasing indirect emissions associated with irrigation and fertilisation events. These findings challenge the conventional emphasis on soil emissions alone as the primary target for climate change mitigation strategies in croplands (Paustian et al., 2016). However, as SOC sequestration is a long-term process and may take years to achieve measurable results, we urge caution when evaluating the climate-smart potential of such practices—especially in short-term assessments. Future studies should account for the temporal dynamics of SOC accumulation when assessing mitigation outcomes.

4.2. *Climate-adaptive optimal crop calendars over time*

Climate change has significantly impacted agricultural production in the NCP. In response, crop planting areas and crop phenology have shifted over the past decades (Gu et al., 2024; Piao et al., 2019). Crop migration, as a crucial climate change adaptation strategy, has been shown to exacerbate environmental pressures, including increased land and water use, energy consumption and GHG emissions (Lam and Chen, 2024). To effectively mitigate these unintended consequences, adjusting crop calendars, as essential technical approaches, must be promptly and appropriately adjusted, ensuring the realisation of climate-smart agriculture. For winter wheat, rising temperatures delay the onset of the dormancy stage while promoting earlier green-up (Xiao et al., 2013). For summer maize, warmer conditions result in earlier flowering and maturity (Xiao et al., 2016). These phenological records support our recommendation to advance top-dressing dates, as timely fertiliser application better synchronises with the crops' accelerated growth phase, thereby promoting crop yield formation. Moreover, improved N use efficiency minimises N losses to the environment, thereby effectively reducing GHG emissions. We also recommend optimising sowing and harvesting dates for summer maize to better allocate more available water, light and thermal resources to the maize growing season. Although shortening the wheat growth period to prioritize maize results in reduced wheat yields (Fig. S4), this trade-off is acceptable as the total annual yields of wheat and maize are significantly increased. Furthermore, local farmers tend to favour maize season due to its higher profitability compared to wheat.

Timely adaptation and implementation of level 3 practices can reduce annual N fertiliser use, irrigation water use and residue inputs by 17.2 %, 6.7 % and 20.0 %, respectively (Fig. 3). Compared with previous regional-scale optimisation studies in the NCP for achieving climate-smart agriculture that did not consider crop calendars (Xiao et al., 2024), our findings indicate that the optimised agricultural inputs in our study are comparable with the regional average recommended values of 358 kg ha^{-1}, 306 mm $year^{-1}$, and 76.5 % for residue retention $year^{-1}$ from Xiao et al. (2024). However, the incorporation of crop calendar adaptation, such as that projected for the 2030s under the SSP245, was found to increase maize yield by 19.1 % and total annual yield by 600 kg ha^{-1}, compared to the previous study. We also found that, unlike current recommendations which allocate more N input and irrigation to the wheat season, our results suggest a more balanced allocation between the wheat and maize seasons. This indicates that incorporating climate-adaptive crop calendar optimisation in the NCP may not lead to more agricultural input savings and net GHG emissions compared to studies that do not include it. However, it can provide greater economic benefits to farmers through increased annual yields, particularly for maize. While local farmers occasionally apply additional irrigation during extreme drought, we recommend the implementation of one additional irrigation event for the maize-growing season. However, this recommendation is subject to local water availability and local government scheduling. Additionally, due to residue retention, the soils at the study site function as C sinks from 1990 to 2100, although sequestration rates decrease over time as they approach the saturation point. The residue retention rate should be further reduced when the SOC sequestration approaches the saturation, since its ability to offset C emission would be diminished. If residue retention is not adjusted in a timely manner, excessive residue retention may have adverse effects such as stimulating soil N_2O emissions, increasing the pest and disease pressures and affecting the seed germination (Wang et al., 2023).

By embedding advance computing techniques into process-based modelling, our study demonstrates how a hybrid modelling approach, combining the DNDC model, LCA and the NSGA–III algorithm, can overcome limitations of traditional single-objective or static simulations in agricultural management. This framework provides a flexible set of trade-off solutions, enabling adaptive and climate-resilient crop calendar design through the simultaneous optimisation of multiple outcomes, including wheat and maize yields, SOC and N_2O emission. Compared to earlier studies(Wang et al., 2022; Xiao et al., 2024; Yan et al., 2024), the use of NSGA–III enabled higher-dimensional optimisation and offered a more comprehensive evaluation of trade-offs and synergies among productivity, environmental impact, and resource efficiency. Overall, this modelling framework holds strong potential for broader application in other cropping systems and regions, providing a scalable tool for designing agriculture under climate change and socio-economic conditions. In addition, optimised crop calendars can deliver both environmental and productivity benefits in the wheat–maize system under climate change. However, realising these benefits requires coordinated efforts from both farmers and policymakers. Targeted training, technical support, and appropriate incentives are essential to promote widespread adoption of these climate-smart practices.

4.3. *Limitations and uncertainties*

In our climate-crop modelling, uncertainties mainly arise from process-based models and climate scenarios (Wang et al., 2020). Specifically, the choice of GCM and the downscaling approaches (e.g. change factor method and dynamical downscaling) represent primary sources of uncertainties in climate predictions (Wang et al., 2024b). The integration of these downscaled climate data into process-based models, combined with the complex and non-linear physiological interactions among crop, soil and climate conditions across various models, further amplifies the uncertainty. In this study, we used locally validated climate and process-based models to ensure more reliable projections to some extent. However, employing ensemble of multiple process-based models and incorporating additional GCMs may be necessary in future research to better capture the responses to climate change (Li et al., 2023; Müller et al., 2017).

Some limitations remain in our simulations. Firstly, although the field study on which our simulations are based was conducted over a 30-year period, we do not account for the potential heterogeneity in soil texture and climate conditions. Addressing such variability typically require data from multiple study sites across the region. Future experiments should aim to collect such data to verify the robustness of the optimal practices. Second, future research should focus on upscaling this approach to derive regional applicable optimisation strategies. Third, due to data limitations, certain additional practices are not considered in this study. However, they need to be explicitly explored in future studies, such as the impact of crop varieties, gene-driven breeding (Chen et al., 2019), the control of extreme climates and the use of enhanced efficiency fertilisers (Lam et al., 2022), etc.

5. Conclusions

Crop calendars were optimised and adapted to achieve climate-smart agriculture in the NCP for both the present and future, using a hybrid model which incorporated the DNDC model, LCA and NSGA–III algorithm. We found that only adjusting N fertiliser rates was insufficient to achieve these goals, while adaptive management of crop calendars was shown to be effective. Specifically, the optimal practice combinations were identified as delaying and reducing basal fertilisation while advancing top-dressing in wheat and both events in maize; postponing wheat sowing and advancing maize sowing; aligning irrigation event with fertilisation, and adding one irrigation event during the maize bell stage; and lowering residue retention. These optimal practices, compared to historical practices, increased crop yields by up to 5.6 % and reduced net GHG emissions by up to 9.4 %, while also decreasing N inputs, irrigation water use and residue inputs by 17.2 %, 6.7 % and 20.0 %, respectively, from 1990 to 2100.

Although integrating climate-adaptive crop calendar optimisation into the NCP may not yield greater input agricultural input savings or net GHG reductions than previous optimisation studies without it, it offers notable economic gains—particularly for maize—by boosting annual yields and significantly mitigating future climate-induced yield losses. Given the minimal yield differences but significant GHG emission disparities across practice levels, trade-offs between incorporating additional practices and the need for further reduction of N inputs and GHG emissions are common and need to be addressed based on a local basis. We found that soil carbon sequestration and the reduction of indirect emissions associated with fertilisation and irrigation were two key contributing factors to the decrease in net GHG emissions resulting from crop calendar adaptation. It is noted that these optimal practices and their effects are climate-adaptive and change over time. Overall, the approach developed in this study and prospects for incorporating adjusting crop calendars into management optimisation can secure food production while improving environmentally friendly and climate-smart agriculture.

Intellectual property

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

We confirm that the manuscript has been read and approved by all named authors. We confirm that the order of authors listed in the manuscript has been approved by all named authors.

Contact with the Editorial Office.

CRediT authorship contribution statement

Deyao Liu: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Baobao Pan:** Writing – review & editing, Validation, Methodology, Conceptualization. **Huarui Gong:** Data curation, Conceptualization. **Jing Li:** Supervision, Funding acquisition, Data curation, Conceptualization. **Enli Wang:** Writing – review & editing. **Jinxi Zhao:** Software. **Yan Xu:** Funding acquisition. **Shu Kee Lam:** Writing – review & editing, Supervision, Funding acquisition. **Deli Chen:** Writing – review & editing, Supervision, Funding acquisition.

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Declaration of competing interest

No conflict of interest exists in the submission of this manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.agsy.2025.104626.

Data availability

Data will be made available on request.

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